Multi-Scale Signal Feature Fusion and Autoregressive Decoding for Event Classification in Distributed Acoustic Sensing

***Abstract*—Distributed acoustic sensing (DAS) uses optical fibre to acquire vibration records over extended monitoring ranges. Event classification requires extracting informative signal features while accounting for relationships among monitoring-point labels. This study presents GC-Transformer, a conditional label-sequence classification model that combines convolutional feature extraction with autoregressive decoding. Waveform and frequency-domain branches process the phase records and the real part of their fast Fourier transform (FFT), respectively. Multi-scale convolutional branches aggregate point-sequence context through additive fusion. The decoder combines causal self-attention with convolution-enhanced cross-attention to predict successive labels. Variable-length sequences are accommodated through padding, with padded targets excluded from the loss and evaluation metrics. The model is trained using teacher forcing and generates labels autoregressively during testing. On the independent test subset, GC-Transformer achieves an accuracy of 98.53% and a macro F1 score of 98.32%, exceeding the strongest listed baseline by 8.93 and 8.86 percentage points, respectively. These results support the use of combined waveform features and conditional label modelling for event classification in distributed optical-fibre acoustic sensing.**

***Index Terms*—Distributed acoustic sensing, Optical-fibre sensing, Event classification, Convolutional neural networks, Autoregressive decoding**

I. Introduction

Distributed acoustic sensing (DAS) uses optical fibre as the sensing medium to acquire vibration signals over tens of kilometres. Interpreting these distributed optical-fibre measurements requires signal-processing methods that distinguish event signatures in the recorded data. By generating high-throughput spatio-temporal monitoring data streams, DAS has been widely applied in industrial monitoring, perimeter security, and urban underground-space safety [1]–[4]. It has also demonstrated considerable potential in seismic exploration and geophysical research [5], [6], as well as in safety monitoring of oil and gas pipelines [7]. Moreover, existing communication fibre networks can be used to construct large-scale sensing systems, further extending the application boundaries of DAS [1].

Despite these advantages, DAS signals exhibit three major characteristics: extremely long sequences, strong background noise, and highly sparse target events. As complex non-stationary signals, they pose substantial challenges for accurate recognition [3], [8]. In practical applications, multi-source interference in complex environments may submerge target signals and reduce recognition accuracy [3]. Meanwhile, high-dimensional noisy data make it difficult for conventional signal-processing methods to extract effective features [1], [2]. In particular, signal distortion under low signal-to-noise ratio conditions severely limits the performance of DAS-based systems in real-world scenarios. Signal-processing techniques for distributed fibre-optic sensing systems, including denoising, feature extraction, and event detection, have therefore been systematically investigated [9]. To address denoising preprocessing for such non-stationary signals, various methods have been proposed. For example, Xu et al. [10] proposed a denoising method based on permutation entropy-optimised variational mode decomposition combined with wavelet thresholding, which effectively improved the signal-to-noise ratio of interferometric fibre-optic vibration signals and provided a foundation for subsequent accurate recognition.

Traditional shallow machine-learning methods rely heavily on manually designed features and struggle to represent high-dimensional non-linear data such as DAS signals, resulting in limited recognition accuracy and low detection efficiency [1], [8]. To overcome these limitations, deep-learning-based event-recognition methods have been introduced into DAS signal analysis, enabling discriminative features to be learned directly from monitoring data [11]. Among existing deep learning methods, convolutional neural networks (CNNs) are effective in capturing local mutations and short-term patterns. However, representing long-range dependencies requires an appropriate receptive field and architecture [12], [13]. Transformer architectures, by contrast, can capture global dependencies through self-attention mechanisms, but a single Transformer architecture is relatively weak in perceiving subtle local fluctuations, and some models suffer from structural complexity and high computational cost [12], [13]. In addition, most existing methods are designed for specific scenarios and have insufficient generalisation capability for cross-scenario multi-class signal classification [3]. These considerations motivate jointly modelling signal features and relationships among neighbouring monitoring-point labels. Researchers have explored the collaborative use of CNNs and Transformers in sensing tasks across different domains. Wu et al. [14] achieved effective fusion of local and global information in hyperspectral and LiDAR land-cover classification by designing feature-coupling units. Lian et al. [15] constructed a multi-scale time-frequency collaborative Transformer for aeroengine fault diagnosis to address the non-stationarity and multi-scale characteristics of vibration signals. In industrial product surface anomaly detection, where anomalous samples are often scarce, Zhang et al. [16] proposed an unsupervised method based on contrastive

learning generative adversarial networks (CLGAN), which improves detection performance by maximising the mutual information between input image patches and reconstructed image patches.

This paper investigates DAS event classification using GC-Transformer, a conditional label-sequence classifier for groups of optical-fibre monitoring-point records. Its waveform and frequency branches extract point-level features, multi-scale convolutional blocks aggregate sequence context, and a conditional decoder predicts successive labels. The experiments compare GC-Transformer with six baseline models on a held-out test subset using accuracy and macro F1 score.

II. Background on DAS signal classification

A. Distributed optical-fibre acoustic sensing

Distributed acoustic sensing (DAS) is a sensing technique that uses optical fibre itself as the sensing medium to detect external mechanical vibrations, acoustic waves, and strain in a spatially distributed manner. Its basic principle relies on Rayleigh backscattering or phase-sensitive optical time-domain reflectometry, whereby external disturbances are inferred from changes in the phase or amplitude of the backscattered light along the fibre [17], [18]. In addition, scattering-enhanced optical fibres have been investigated to improve the backscattered signal strength and sensing performance of DAS systems [19].

Compared with conventional point sensors, DAS can virtualise tens of kilometres of optical fibre into thousands of continuous acoustic sensing channels. It therefore offers several advantages, including long-distance coverage, resistance to electromagnetic interference, and continuous all-weather monitoring. Owing to these characteristics, DAS has been widely used in perimeter security, oil-pipeline leakage monitoring, railway safety inspection, and seismic-wave acquisition [4], [5].

Nevertheless, data collected by DAS systems are large in scale and highly complex. DAS signals are usually represented as spatio-temporal matrices, which contain not only the signatures of target events but also substantial environmental background noise and fading noise from the sensing system itself. The signal characteristics of coherent-detection-based distributed fibre-optic vibration sensing systems have also been investigated, showing that system-level sensing characteristics must be considered during subsequent signal processing [20]. In practical engineering environments, DAS signals are often affected by the strong coupling between system fading noise and complex environmental background noise. Together with the transient and sparse nature of target events in the time domain, this low signal-to-noise ratio makes it difficult for conventional signal-processing methods based on energy thresholds or shallow statistical features to achieve high-accuracy recognition. Therefore, developing a data-driven feature-extraction framework that can adapt to complex noisy environments remains a key challenge in advancing DAS towards intelligent sensing.

B. Convolutional feature extraction

CNNs, originally developed for image and grid-structured data processing, have increasingly been applied to one-dimensional time-series signal analysis because of their ability to learn local patterns directly from raw signals [21]. In DAS signal processing, their main advantages arise from local receptive fields and weight sharing, which allow the network to capture locally translation-invariant patterns. Existing studies commonly feed one-dimensional vibration sequences directly into a 1D-CNN, or first convert the signals into time-frequency representations using the short-time Fourier transform and then process them with a 2D-CNN. Similar developments have been reported in vibration-based structural damage detection, where machine-learning and deep-learning methods have progressively replaced or complemented conventional signal-processing approaches [22]. These applications motivate the use of convolutional feature extraction for local waveform and time-frequency patterns in DAS signals [11], [21].

The context available to a convolutional layer is determined by its receptive field. Bai et al. investigated convolutional sequence models and demonstrated that appropriately designed temporal convolutional networks can capture long-range dependencies [23]. Consequently, the ability of a CNN to represent extended signal patterns depends on its architecture, including kernel sizes, dilation, and depth. In DAS monitoring, local feature extraction should therefore be complemented by an explicit mechanism for aggregating context across the input sequence.

To overcome this limitation, researchers have introduced Transformers into one-dimensional time-series analysis. The self-attention mechanism of the Transformer can directly model dependencies between arbitrary time steps in a sequence, without relying solely on deeper layer stacking to enlarge the receptive field. This property supports the representation of global temporal relationships in sequence-modelling tasks [24]. Transformer-based encoders have also been studied for multivariate time-series representation learning [25]. Therefore, combining the local feature-extraction capability of CNNs with the global modelling capability of Transformers provides an effective approach for processing complex DAS signals.

C. Transformers and Generative Models

To address the difficulty of long-sequence modelling, Vaswani et al. introduced the Transformer architecture, which changed the paradigm of sequence modelling through the self-attention mechanism. Unlike CNNs, which are constrained by local receptive fields, self-attention allows direct weighted interactions between any two time steps in a sequence. As a result, global dependencies can be captured in parallel within a fixed number of computational steps. The effectiveness of this architecture has also been demonstrated in time-series analysis. For example, the TST model proposed by Zerveas et al. showed that Transformer-based encoders can extract more robust semantic representations from multivariate time series than LSTM- and CNN-based models [25].

Nevertheless, a standalone Transformer still has limitations in modelling fine-grained local details. To take advantage of both local information and global dependencies, hybrid architectures have increasingly been explored. For example, Gulati et al. proposed Conformer for speech recognition, combining convolutional modules with self-attention to model local and global dependencies [26]. This idea is relevant to DAS signal processing, because DAS signals often contain both high-frequency transient details and long-term evolutionary patterns, which are difficult to capture adequately with a single architecture.

The development of large-scale autoregressive language models has demonstrated that sequence models can exploit contextual dependencies and adapt to downstream tasks from limited examples [27]. Meanwhile, conditional autoregressive modelling has shown potential beyond conventional language generation. Rather than predicting all labels independently, an autoregressive classifier factorises the conditional probability of a label sequence as $P(Y \mid X) = \prod_{s=1}^{S} P\left(y_s \mid y_{<s}, X\right)$. Each label is therefore predicted using both the input signal and the previously generated labels. This formulation allows dependencies between adjacent labels to be explicitly incorporated into the decision process.

Generative models have also been investigated for DAS event classification in telecom networks [28]. That work motivates examining generative decision rules for sensing tasks. The present study focuses on autoregressive label-sequence generation conditioned on multi-scale signal features.

Therefore, integrating the local feature-extraction capability of CNNs, the global dependency-modelling capability of Transformers, and a generative autoregressive strategy provides an effective technical basis for complex time-series analysis of DAS signals. It also provides a theoretical basis for the design of the CNN–Transformer generative classification model proposed in this study.

III. Autoregressive convolutional transformer for DAS event classification

To address the non-stationarity and label noise of large-scale DAS signals under complex operating conditions, this study proposes GC-Transformer, a generative CNN-Transformer hybrid model. Unlike conventional discriminative classification methods, GC-Transformer reformulates time-series signal classification as a signal-to-sequence generation problem. This design enables the model to capture fine-grained local features while making use of global contextual information. The proposed model consists mainly of a multi-scale CNN feature-extraction module, a Transformer-based context representation, a generative decoder, and a dynamic masking decision head. The design and implementation of these modules are described in the following subsections, and the overall workflow is illustrated in Fig. 1.

A. Overall Model Architecture

GC-Transformer follows an encoder-decoder architecture, as illustrated in Fig. 1. The input is a DAS tensor $X \in \mathbb{R}^{B\times S\times C\times L}$, where B is the batch size, $S \leq 9$ is the number of valid monitoring-point records in a sample, $C = 1$ is the raw phase channel, and $L = 6{,}144$ is the waveform length per point. Samples group records from the same observation interval. The loader retains the filesystem enumeration order within each observation group and draws contiguous subsequences of that list. Consequently, sequence positions describe the order of the loaded records and are not guaranteed to correspond to consecutive spatial positions along the fibre. The target is $Y = (y_1, \ldots, y_S)$, with one class label per valid monitoring point. Each waveform spans approximately 2.46 s at a sampling frequency of 2,500 Hz.

The main functional components are described below:

The signal encoder maps each monitoring point to a feature vector using waveform and frequency-domain convolutional branches. The latter processes the real part of the full-length discrete Fourier transform. Both branches use three one-dimensional convolutions with average and max pooling. The final convolutional transformation is shared between the branches. Their outputs are added, passed through ReLU, flattened, and projected to the hidden dimension. Four multi-scale convolutional blocks then aggregate information across the point-record sequence.

The decoder incorporates fixed sinusoidal positional encoding and masked self-attention over the target labels. It receives point-level features as memory and predicts the class at the next label position. The implementation stores four references to one shared decoder block and calls each reference with the same target input and memory; only the final call's output is returned. This computation uses one shared block rather than four successively stacked, independently parameterised blocks.

The dynamic masking decision head maps the decoder output to the class space through linear projection and uses a dynamic padding strategy to handle variable-length sequences, producing the final classification sequence $Y = (y_1, y_2, \ldots, y_S)$.

This design combines local waveform features and sequence context for DAS event recognition.

B. Multi-scale CNN Feature-Extraction Module

The waveform and frequency-domain branches produce a feature vector for each monitoring point. Multi-scale convolutional blocks operate on this point-level feature sequence, as illustrated in Fig. 2.

Let $Z \in \mathbb{R}^{B\times S\times D}$ denote the input point-level features, where D is the hidden dimension. For h feature groups, Z is reshaped to $\widehat{Z} \in \mathbb{R}^{B\times h\times S\times d_h}$, where $d_h = D/h$. Padded positions are set to zero in the convolutional branch input $\widehat{Z}_{\mathrm{mask}}$. Four convolutional branches use kernel sizes $(k, 1)$, with $k \in \mathcal{K} = \{1, 3, 5, 7\}$, to aggregate features across the monitoring-point sequence. Each branch is computed as in (1):

$$H_k = \mathrm{ReLU}\Big(\mathrm{Conv}_{(k,1)}\Big(\mathrm{Dropout}(\widehat{Z}_{\mathrm{mask}})\Big)\Big), \quad k \in \mathcal{K}. \tag{1}$$

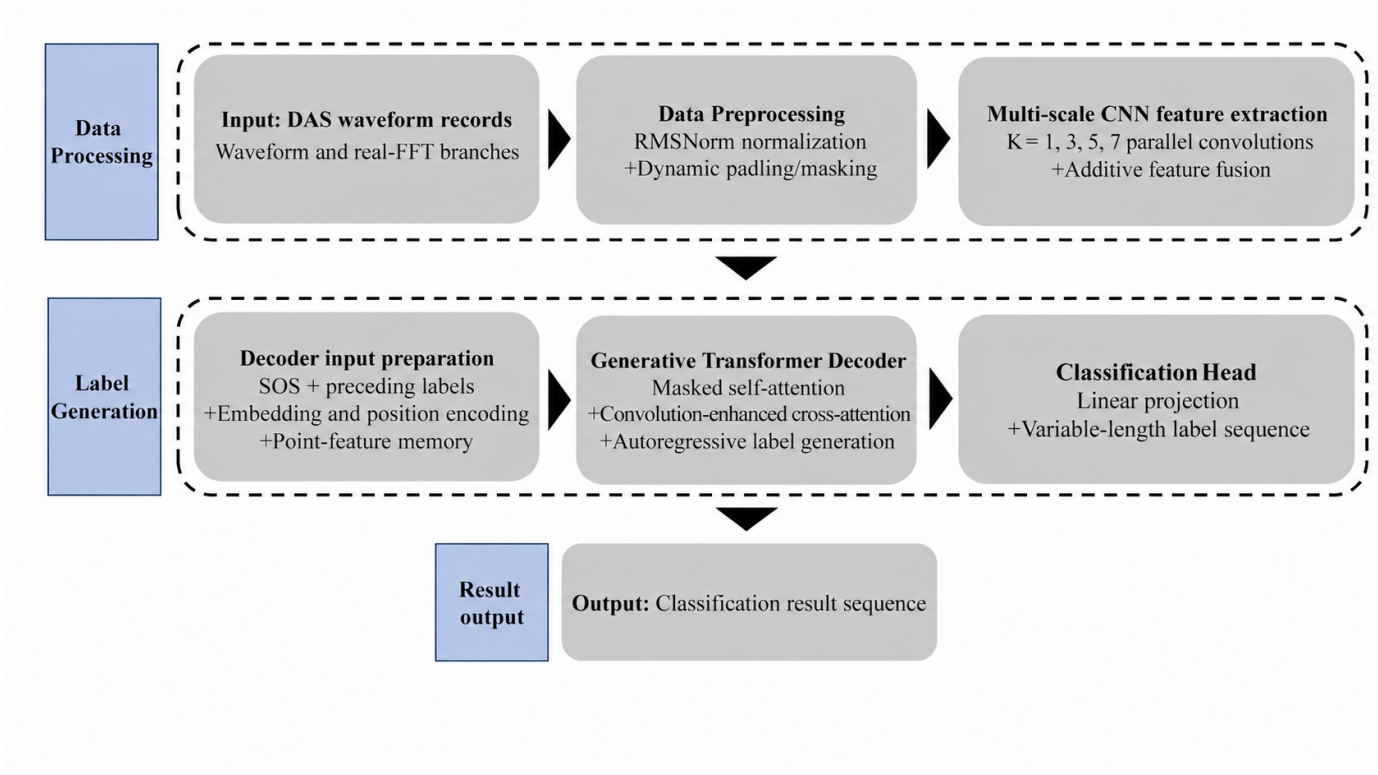


Fig. 1. GC-Transformer combines waveform and real-FFT features with conditional label-sequence generation. Waveform and real-FFT branches provide multi-scale point-sequence features as decoder memory. The decoder combines embedded label history with masked self-attention and convolution-enhanced cross-attention. A classification head projects decoder outputs to the class space.

Padding of $(k-1)/2$ along the sequence dimension preserves the number of monitoring positions. The four branch outputs are added, followed by dropout and a residual connection with $\widehat{Z}$. After reshaping back to $B \times S \times D$, RMSNorm is applied. A two-layer feed-forward module with ReLU and dropout is then followed by a second residual connection and RMSNorm. Thus, the multi-scale block aggregates information across neighbouring positions in the loaded feature sequence, whereas the preceding one-dimensional convolutions extract features within each waveform.

C. Convolution-Enhanced Attention

To support label-sequence generation, each decoder block combines masked multi-head self-attention with convolution-enhanced cross-attention and a feed-forward module. Residual connections and RMSNorm are applied around the three processing stages. The self-attention stage uses linearly projected queries, keys, and values with a causal attention mask. The cross-attention stage enhances the query representation using multi-scale convolutions; its keys and values are separate linear projections of the point-feature memory.

For the cross-attention operation illustrated in Fig. 3, let H_{dec} denote the decoder hidden states and M_{enc} the point-feature memory. The query, key, and value representations are formed as in (2):

$$\begin{aligned} Q_0 &= \mathrm{SplitHeads}(\mathrm{Linear}_Q(H_{\mathrm{dec}}))\,, \\ Q &= Q_0 \\ &\quad + \mathrm{Dropout}\left(\sum_{k\in\mathcal{K}} \mathrm{CausalConv}_{(k,1)}(Q_0)\right), \\ K &= \mathrm{SplitHeads}(\mathrm{Linear}_K(M_{\mathrm{enc}}))\,, \\ V &= \mathrm{SplitHeads}(\mathrm{Linear}_V(M_{\mathrm{enc}}))\,. \end{aligned} \tag{2}$$

Here, $\mathcal{K} = \{1, 3, 5, 7\}$, and SplitHeads reshapes a hidden vector into h attention groups of width d_h. For a query convolution of size k, padding by $k-1$ positions is followed by removal of the final $k-1$ outputs. The retained output at a position consequently depends on that position and preceding query positions. The resulting query is used with K and V in scaled dot-product attention, with scale $\sqrt{d_h}$. Attention-head outputs are merged and transformed by a two-layer feed-forward mapping.

At decoding position s, the target input contains the start symbol and preceding labels. The causal self-attention mask and causal query convolutions preserve this label ordering. Fixed sinusoidal positional encoding represents the positions of the target tokens. Label generation models dependencies between monitoring-point classes rather than temporal evolution within an individual waveform.

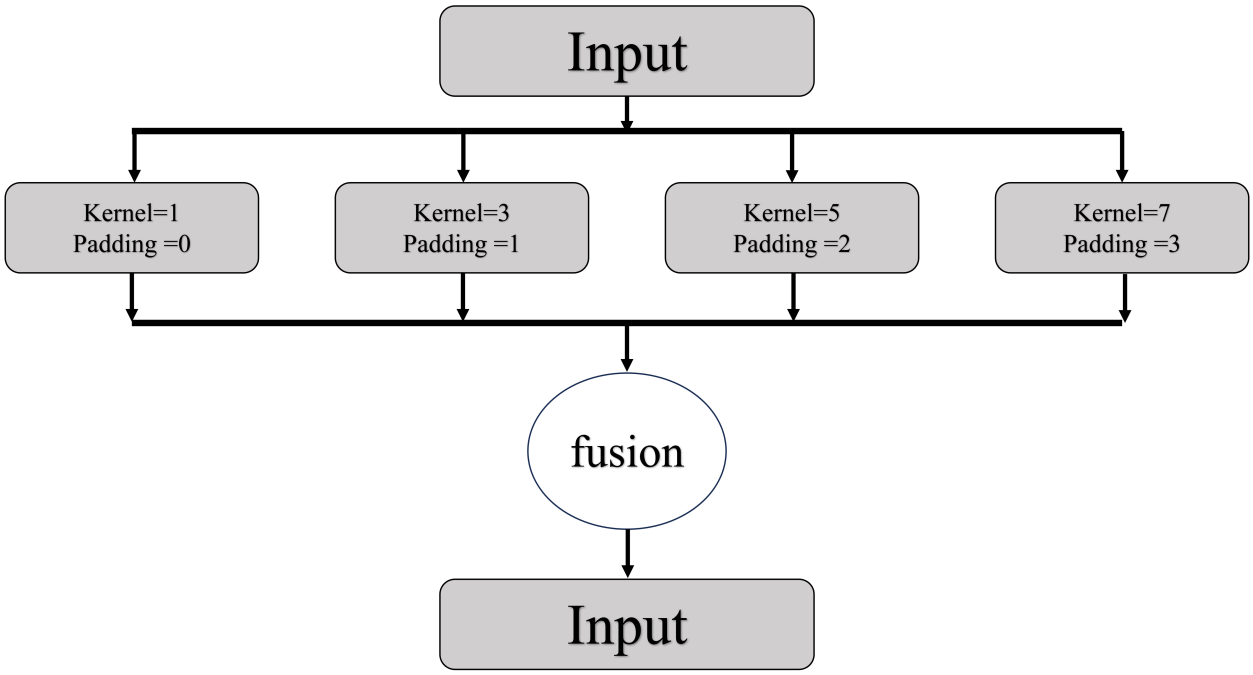


Fig. 2. Multi-scale convolutional block for point-level features. Branches with kernel sizes 1, 3, 5, and 7 aggregate information across sequence positions. The branch outputs are summed and combined with a residual connection.

D. Classification Mechanism Based on Generative Decoding

The main idea of GC-Transformer is to reformulate conventional discriminative classification as autoregressive generative classification. During training, teacher forcing is used to feed the original signal X and the ground-truth label sequence Y into the model, with a start-of-sequence token added to the beginning of the label sequence. The training loss averages the negative log-likelihood over valid labels within each sample and then over the batch, as in (3), where S_b is the number of valid labels in sample b:

$$\mathcal{L}_{\mathrm{CE}} = -\frac{1}{B}\sum_{b=1}^{B}\frac{1}{S_b} \times \sum_{s=1}^{S_b}\log P(y_{b,s} \mid y_{b,<s}, X_b; \theta)\,. \tag{3}$$

Teacher forcing supplies the start token followed by the ground-truth labels to the decoder. The output at position $s-1$ is compared with label y_s, so that the target labels are shifted by one position relative to the supervised outputs. The five class labels use token IDs 0–4; the start token uses ID 5. Padding targets are marked -1 and mapped to embedding ID 6 for the decoder input. Loss is computed over the valid label prefix of each sample.

During testing, decoding starts from [SOS]. The model selects the highest-scoring next class and feeds that prediction back to generate subsequent labels. Predictions and reference labels are accumulated across test batches, and padded positions are excluded from the evaluation metrics.

E. Variable-Length Sequence Handling Strategy

In industrial monitoring, the sequence length of DAS signals is often variable. To improve the generalisation capability of the model, this study designs a dynamic padding and masking strategy, followed by logical alignment in the post-processing stage.

Dynamic padding: For segments shorter than the maximum point-sequence length $S_{\max}$, zeros are appended to the end of the signal, while ignore indices are appended to the end of the label sequence. This ensures consistency during batch processing.

Mask filtering: Padding positions are identified using the ignore label. The point-level convolutional blocks use this mask to zero the padded branch inputs, and padded targets are excluded from loss and metric computation.

Sequence construction: For each observation group, the loader draws five random contiguous slices of its file-enumeration list. Each slice contains between one and nine valid point records and is padded to nine positions. The training entry point applies this sampling procedure to all three observation subsets before partitioning the resulting samples.

Padding accommodates variable numbers of point records and is separate from the random subsequence sampling used to construct samples. The point-feature memory is supplied

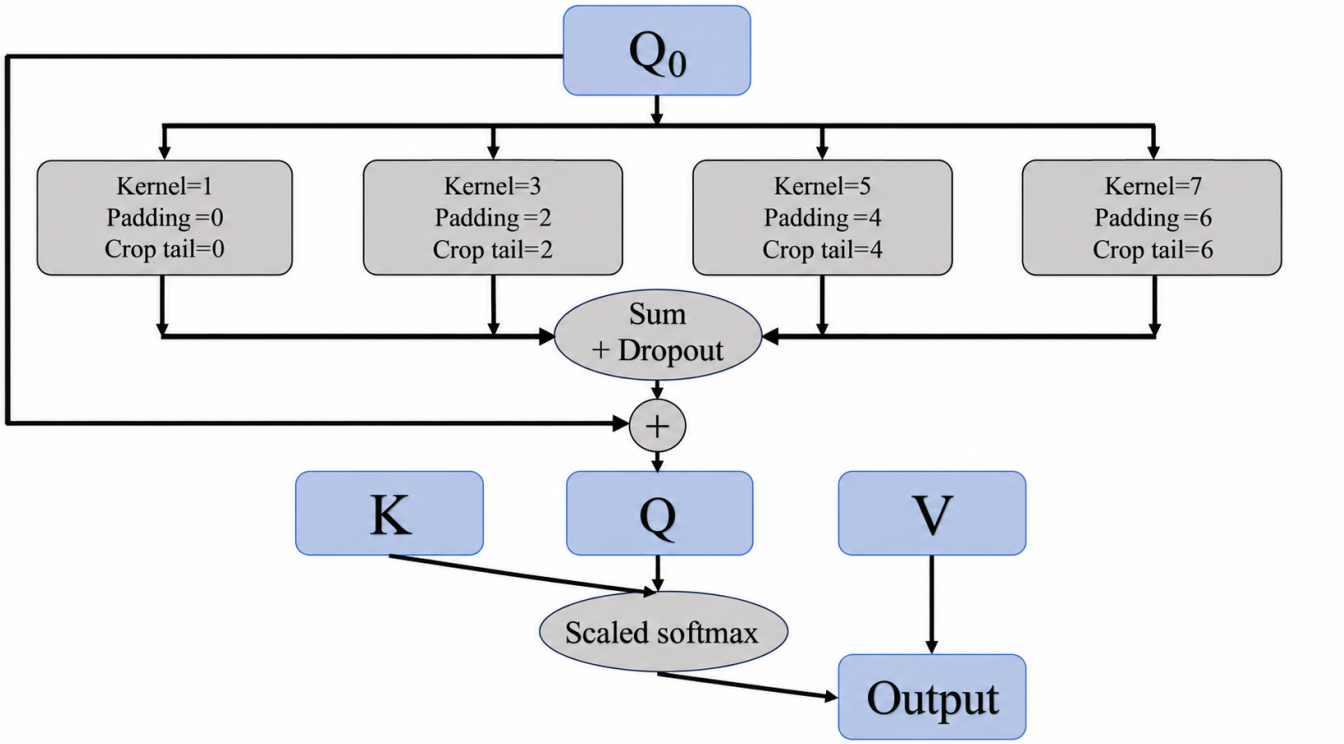


Fig. 3. Convolution-enhanced cross-attention uses causal multi-scale query enhancement. Kernels 1, 3, 5, and 7 use padding 0, 2, 4, and 6, respectively, followed by cropping of the corresponding number of trailing positions. The branch sum passes through dropout and is added to the initial query. Keys and values are linear projections of point-feature memory, as specified in (2).

to cross-attention without a source key-padding mask in this implementation; masking the convolutional branch inputs therefore does not exclude every padded memory position from attention.

IV. Experimental evaluation

A. Experimental Dataset and Evaluation Metrics

The dataset contains phase records from three observation subsets. Their label files contain 4,264, 12,688, and 4,732 point records in 164, 488, and 182 observation groups, respectively: 21,684 records in 834 groups overall. Each group contains labels for 26 monitoring points, numbered 425–450. A binary point record contains 6,144 signed 32-bit values, converted to phase values by division by 2^{22}. The configured sampling frequency is 2,500 Hz. The loader generates five subsequence samples per successfully processed observation group, giving an upper bound of 4,170 samples before partitioning when all 834 groups are available and valid. The exact accepted sample count depends on file availability and validity checks.

During preprocessing, records from the same observation group were read and matched to class labels by point identifier. The loader retains file-enumeration order and samples up to nine records from contiguous positions in that list. Zero padding extends shorter sequences to nine positions, and padded labels use -1. The waveform and real-FFT representations are independently normalised using RMSNorm over their 6,144 values per point. The FFT magnitude plots in Fig. 5 illustrate the signals; the model's frequency branch uses the FFT real part rather than the plotted magnitude.

The five labels, coded 0–4, represent normal background (52.52%), vibrations from pipeline operations (3.43%), other noise (15.62%), subway noise (7.29%), and road-traffic noise (21.14%), respectively. These proportions were calculated from the original records and exclude resampled and padded positions. Labels were assigned manually using waveform shape, relative energy, and vibration patterns at neighbouring points. When other noise coincided with pipeline-operation, subway, or road-traffic signals, the record was assigned to the corresponding operation or traffic class. Overlapping pipeline-operation and traffic signals were classified using waveform features and relative energy. The padding label -1 was excluded from the five-class task.

Accuracy and macro F1 score are calculated from predicted and true labels on the complete test subset after removing padded positions. Macro F1 gives equal weight to each class. GC-Transformer additionally reports macro precision, while the baseline implementations report macro recall. Precision measures the fraction of predictions of a class that are correct, whereas recall measures the fraction of true instances of a class that are detected. The two metrics are presented in separate columns, and cross-model comparisons use accuracy and macro F1.

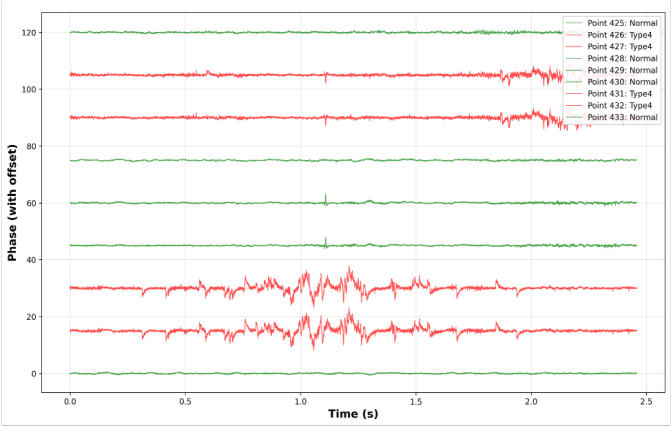


Fig. 4. Time-domain phase signals from nine consecutive monitoring points acquired during the same observation interval. Traces are vertically offset for readability; the legend identifies the monitoring point and its recorded class label. The legend label Type4 denotes road-traffic noise.

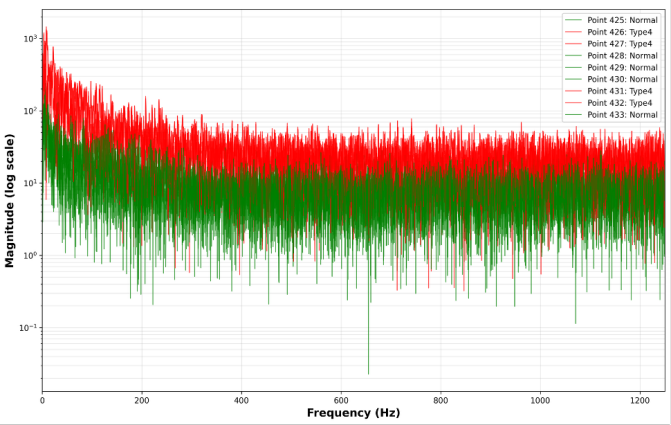


Fig. 5. FFT magnitude spectra of the same nine monitoring points shown in Fig. 4. The frequency range is 0–1,250 Hz, and magnitude is displayed on a logarithmic scale.

B. Experimental Setup

This study uses a sequence-to-sequence model that combines CNN-based feature extraction with a Transformer decoder for sequence classification of optical-fibre phase data. The model input consists of a sequence of at most nine monitoring points, where each point corresponds to a phase sequence of length 6,144. The feature-extraction module contains three one-dimensional convolutional layers with 32, 64, and 64 channels, respectively. Average pooling and max pooling are used for downsampling. In addition, time-domain and frequency-domain branches are fused, and RMSNorm normalisation is applied to improve the representation of anomaly patterns.

The hidden dimension is 256. Four independent point-level convolutional blocks use eight feature groups. The decoder block uses eight attention heads, fixed sinusoidal positional encoding, and a dropout probability of 0.5. Its four configured entries reference the same block and repeatedly process the original target input, as described above.

After subsequence construction, samples are randomly partitioned into training, test, and validation subsets in proportions of 60%, 20%, and 20%, using two successive random train–test splits. The code does not group samples by observation interval or event during partitioning and does not fix a random seed. Different samples drawn from one observation group can therefore be assigned to different subsets. This partitioning evaluates held-out samples from the sampled pool rather than held-out observation periods or monitoring sites.

The Adam optimiser uses a learning rate of 1×10^{-4}, a weight decay of 1×10^{-5}, and a batch size of 64. The configuration specifies 1,000 epochs. Training, test, and validation metrics are calculated after each epoch. The model is saved after the final epoch; the script does not select a checkpoint using the best validation metric. The implementation uses PyTorch and requests CUDA execution when available.

All three observation subsets are loaded with random subsequence sampling enabled before the train–test–validation split. Consequently, validation and test samples also originate from the resampled pool. The function named "amplify_audio" returns the original phase values after validity checks, so it applies no gain amplification. Padding positions are excluded from the supervised loss and label statistics.

C. Comparative Evaluation

To comprehensively evaluate the effectiveness of the proposed method for distributed optical-fibre sensing signal classification, we conducted systematic comparisons with several representative existing methods. The comparison models were divided into two groups: conventional convolutional neural network (CNN) models and representative methods for audio and temporal signal modelling.

In the first group, four widely used CNN architectures were selected, namely ResNet [29], InceptionNetV4 [30], ShuffleNetV2 [31], and MobileNetV3 [32]. These models have been extensively applied in image and time-series signal modelling tasks. They represent deep residual learning, multi-scale convolutional feature extraction, lightweight channel-shuffling networks, and efficient mobile-oriented architectures, respectively. These models provide representative convolutional baselines for this task.

In the second group, two representative methods from audio understanding were introduced: Audio Spectrogram Transformer (AST) [33] and BAST-Vanilla [34]. AST is a pure Transformer-based audio spectrogram modelling method that captures long-range temporal dependencies through self-attention. BAST-Vanilla uses the vanilla Transformer variant of the BAST framework, with shared branch parameters and subtraction-based integration. These methods provide Transformer-based comparison architectures.

GC-Transformer is compared with six baseline models. Accuracy and macro F1 are common metrics; the GC-Transformer script computes macro precision, whereas the baseline scripts compute macro recall. ResNet and InceptionNetV4 are custom residual and Inception-style implementations. ShuffleNetV2 uses the 1.0-width torchvision variant, and

MobileNetV3 uses the Small variant, both without pretrained weights; their waveform inputs are interpolated to 224 by 224 and mapped to three channels. AST uses the base224 configuration without ImageNet or AudioSet pretraining, reshaping waveform values to a 48-by-128 input. The BAST-Vanilla adapter reshapes each waveform to 128 by 48, duplicates it into two input branches, and uses shared branch parameters with subtraction-based integration.

D. Comparative Results

Table I presents the results on the independent test subset. GC-Transformer achieves an accuracy of 0.9853, a macro precision of 0.9800, and a macro F1 score of 0.9832. Accuracy and macro F1 are reported for all models. Macro precision and macro recall are displayed separately; unavailable values are marked with a dash and are excluded from cross-model comparisons.

1) Comparison of Classification Performance: GC-Transformer achieves the highest accuracy and macro F1 score among the compared models. Its accuracy and macro F1 score are 0.9853 and 0.9832, compared with 0.8960 and 0.8946 for BAST-Vanilla, the strongest baseline on these two metrics. These differences correspond to improvements of 8.93 and 8.86 percentage points, respectively. Its macro precision is 0.9800; macro precision is reported separately from the baseline macro recall values.

2) Discussion: The implemented architecture combines point-level waveform and frequency features, multi-scale processing of the loaded point-record sequence, and conditional next-label prediction. Convolutional query enhancement and causal self-attention preserve the ordering of label prediction under teacher forcing. Filesystem order determines the loaded sequence in the current data pipeline, so physical adjacency is not guaranteed.

The sample-level random partition allows samples from the same observation group to occur in different subsets. The results therefore assess classification of held-out samples from the sampled pool; they do not establish generalisation to unseen observation periods or monitoring sites. The comparison also does not isolate the contributions of individual modules. Group-based evaluation and module ablation are directions for further work.

V. Conclusions

GC-Transformer combines dual-branch signal feature extraction, multi-scale point-sequence convolutions, and a convolution-enhanced conditional decoder for DAS classification. The model is trained with ground-truth label history and generates labels autoregressively during testing. On the independent test subset, it achieves an accuracy of 98.53%, a macro precision of 98.00%, and a macro F1 score of 98.32%. Accuracy and macro F1 exceed those of the strongest listed baseline by 8.93 and 8.86 percentage points, respectively. These results support combining local signal features with conditional label-sequence modelling. Further work will assess generalisation across observation groups and monitoring sites and quantify the contribution of each model component.

Statements and Declarations

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Competing interests

The authors declare that they have no competing financial or non-financial interests related to this work.

Author approval

All authors have read and approved the manuscript and agree to its submission to DSIS 2026.

AI-assisted figure preparation

OpenAI image-generation tools were used to assist in revising the schematic diagrams in Figs. 1 and 3. These diagrams illustrate the model architecture rather than experimental measurements. The authors retain responsibility for the accuracy and final content of the figures.

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TABLE I

CLASSIFICATION RESULTS ON THE INDEPENDENT TEST SUBSET. VALUES ARE FRACTIONS. GC-TRANSFORMER REPORTS MACRO PRECISION, AND THE BASELINES REPORT MACRO RECALL. DASHES DENOTE UNAVAILABLE VALUES. ACCURACY AND MACRO F1 PROVIDE THE COMMON COMPARISON METRICS.

Method	Accuracy	Macro precision	Macro recall	Macro F1
ResNet [29]	0.8690	–	0.7604	0.7974
InceptionNetV4 [30]	0.8719	–	0.8147	0.8359
ShuffleNetV2 [31]	0.8582	–	0.7507	0.7843
MobileNetV3 [32]	0.8476	–	0.7433	0.7766
AST [33]	0.8872	–	0.9362	0.8925
BAST-Vanilla [34]	0.8960	–	0.8824	0.8946
Ours	**0.9853**	**0.9800**	–	**0.9832**

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